worksheet # 06

Date: Lec # 08 Day: M T W T F S

Clairaut's theorem :-

Suppose f is defined on a Disk (Domain) D that contains the point (a, b). If the function f_{xy} and f_{yx} are both continuous on D, then

$$\boxed{f_{xy}(a,b) = f_{yx}(a,b)}$$

Exp:- Verify the conclusion of Clairaut's theorem hold that is

$$\boxed{u_{xy} = u_{yx}}$$

(i) $u = x^4y^3 - y^4$

Solution:-

$f(x) = 4x^3y^3$

U_{xy} $f(x,y) = 12x^3y^2$

$f(y) = 3x^4y^2 - 4y^3$

U_{yx} $f(y,x) = 12x^3y^2$

Omit

Hence $U_{xy} = U_{yx}$

Clairaut's theorem satisfies.

$u_{xy} = u_{yx}$

L.H.S = R.H.S

$u(x) = 4x^3y^3 - 0 \quad = \quad u(y) = 3x^4y^2 - 4y^3$

$u(x,y) = 12x^3y^2 \quad = \quad u(y,x) = 12x^3y^2$

Hence $u_{xy} = u_{yx}$

Clairaut's Theorem satisfied

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Date: Day: M T W T F S

Exp :- $f(x,y,z) = \sin(3x + yz)$ find f_{xxyz}.

Solution :-

$$f_x = \cos(3x + yz) \times 3 = 3\cos(3x + yz)$$

$$f_{xx} = -3\sin(3x+yz) \times 3 = -9\sin(3x+yz)$$

$$f_{xxy} = -9\cos(3x+yz) \times z = -9z\cos(3x+yz)$$

$$f_{xxyz} = -9\left[\frac{\partial}{\partial z}\left(z \times \cos(3x+yz)\right)\right]$$

$u = z$

$v = \cos(3x+yz)$

$u' = 1$

$v' = -\sin(3x+yz) \times y$

$$f_{xxyz} = -9\left[(1 \times \cos(3x+yz) + (z \times -y\sin(3x+yz)\right]$$

$$f_{xxyz} = -9\left[\cos(3x+yz) - yz\sin(3x+yz)\right]$$

$$= -9\cos(3x+yz) + 9yz\sin(3x+yz)$$

Ans.

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